

Lab 1:

```
1  WITH RankedFilm AS(  
2      SELECT f.title, f.length, cat.name,  
3      DENSE_RANK() OVER(  
4          PARTITION BY cat.name  
5          ORDER BY f.length  
6      ) as rank_by_length  
7      FROM film f JOIN film_category fc  
8      ON f.film_id = fc.film_id  
9      JOIN category cat ON  
10     fc.category_id = cat.category_id  
11 )  
12 SELECT * FROM RankedFilm WHERE rank_by_length <=3;
```

Data Output Messages Notifications



	title character varying (255) 🔒	length smallint 🔒	name character varying (25) 🔒	rank_by_length bigint 🔒
1	Suspects Quills	47	Action	1
2	Excitement Eve	51	Action	2
3	Lust Lock	52	Action	3
4	Caddyshack Jedi	52	Action	3
5	Side Ark	52	Action	3
6	Doors President	49	Animation	1
7	Hook Chariots	49	Animation	1
8	Champion Flatliners	51	Animation	2
9	Harper Dying	52	Animation	3
10	Labyrinth League	46	Children	1

Lab 2:

```

14 WITH RevenuePerMonth AS (
15     SELECT DATE_TRUNC('month', payment_date) AS payment_month,
16     SUM(amount) as month_revenue
17     FROM payment
18     GROUP BY payment_month
19 )
20 SELECT *,
21 month_revenue - LAG(month_revenue,1) OVER(ORDER BY payment_month) as diff,
22 ((month_revenue - LAG(month_revenue,1) OVER(ORDER BY payment_month))/
23 LAG(month_revenue,1)OVER(ORDER BY payment_month))*100 as growth_rate
24 FROM RevenuePerMonth;

```

Data Output Messages Notifications

Showing rows: 1 to 4

	payment_month timestamp without time zone	month_revenue numeric	diff numeric	growth_rate numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	15534.72	186.0035632866530000
3	2007-04-01 00:00:00	28559.46	4672.90	19.56288389789069669300
4	2007-05-01 00:00:00	514.18	-28045.28	-98.19961581906660700200

Lab 3:

```

27 SELECT customer_id, payment_id, payment_date, amount,
28 SUM(amount) OVER(
29     PARTITION BY customer_id
30     ORDER BY payment_date
31 ) AS clv
32 FROM payment;

```

Data Output Messages Notifications

Showing rows:

	customer_id smallint	payment_id [PK] integer	payment_date timestamp without time zone	amount numeric (5,2)	clv numeric
1	1	18495	2007-02-14 23:22:38.996577	5.99	5.99
2	1	18496	2007-02-15 16:31:19.996577	0.99	6.98
3	1	18497	2007-02-15 19:37:12.996577	9.99	16.97
4	1	18498	2007-02-16 13:47:23.996577	4.99	21.96
5	1	18499	2007-02-18 07:10:14.996577	4.99	26.95
6	1	18500	2007-02-18 12:02:25.996577	0.99	27.94
7	1	18501	2007-02-21 04:53:11.996577	3.99	31.93
8	1	22680	2007-03-01 07:19:30.996577	4.99	36.92

Lab 4:

```

35  SELECT co.country, c.customer_id,
36         CONCAT(c.first_name, ' ', c.last_name) as name,
37         SUM(p.amount) AS total_spend,
38         RANK() OVER(
39             PARTITION BY co.country
40             ORDER BY SUM(p.amount) DESC
41         ) AS spend_rank
42  FROM customer c
43  JOIN address add ON c.address_id = add.address_id
44  JOIN city ci ON add.city_id = ci.city_id
45  JOIN country co ON ci.country_id = co.country_id
46  JOIN payment p ON c.customer_id = p.customer_id
47
48  GROUP BY name, co.country, c.customer_id
49  )
50
51  SELECT * FROM RankedCusCoun
52  WHERE spend_rank = 1;

```

Data Output Messages Notifications



Showing rows: 1 to 108

	country character varying (50)	customer_id integer	name text	total_spend numeric	spend_rank bigint
1	Afghanistan	218	Vera Mccoy	67.82	1
2	Algeria	176	June Carroll	151.68	1
3	American Samoa	320	Anthony Schwab	47.85	1
4	Angola	528	Claude Herzog	93.80	1
5	Anguilla	381	Bobby Boudreau	99.68	1
6	Argentina	560	Jordan Archuleta	129.71	1